

Client Project Proposal & Sprint 1 Plan

IBM × RMIT Capstone — COSC2408 Programming Project 1

Part A — Project Proposal Summary

Proposal prepared ahead of the next client discussion. Our first client meeting (7 Aug 2026) covered the project brief and initial direction; this proposal reflects that discussion and the brief provided. A follow-up meeting with Emily is still needed to formally agree Part A before Part B is locked in, per the assignment instructions.

Project Details

Project name: IBM — AI-Powered Consulting Simulation: Digital Transformation Game 1

Client / organization: IBM (contact: Emily Chin, emily.chin@ibm.com)

Prepared by: Team 1

Date of client discussion: 7 August 2026 (initial); formal Part A sign-off pending follow-up meeting

Team & Roles

Role	Assigned to	Notes
PM	Gayath Wethmin Kaluwahewa	Overall coordination, client/supervisor liaison, Sprint sign-off
BA	Fatima Hubali	Requirements, persona briefs grounded in consultant interviews
UX	Amritha Selvaganapathi	Game world/UI design, prototypes
Dev	Kashaf Fatima	Frontend (Next.js + Phaser), infra
Dev	Ibrahim Allouche	Backend (FastAPI + Anthropic API), infra

Problem Statement

New graduates joining IBM's consulting practice have no real sense of what the job looks like day to day, cold emails that get ignored, meetings that go sideways, and clients who don't know what they want. Right now, those mistakes happen in front of real clients because there's no low-stakes way to practise the full engagement lifecycle first.

Target Users

Trainee/graduate consultants joining IBM's consulting practice who have not yet run a real client engagement.

Proposed Solution

A single-player, RPG-style training simulation set in a 2D office environment. The trainee walks their character up to LLM-powered client and stakeholder personas to run outreach, prepare for meetings, and hold live client conversations that react dynamically with realistic objections and decision-making building toward closing an in-game contract.

Scope Boundaries

In scope this term:

- A playable, web-based 2D simulation covering the full consulting loop: find a lead, run cold outreach, prep for and run a client meeting, close a contract
- At least one LLM-powered client persona that responds dynamically to player input with realistic objections, questions, or pushback
- Game state tracking portfolio progress (contracts closed, relationship health, deal quality) across a session
- Consultant mechanics grounded in real practice shared by IBM consultants (via interviews/shadowing), not invented from scratch

Out of scope this term:

- Multiplayer or native mobile builds
- A full production deployment for IBM's actual training cohorts this term delivers a capstone-quality prototype, not the final reusable enterprise asset
- Voice-acted dialogue
- A large roster of client personas beyond what's needed to demonstrate realistic, dynamic behaviour, we'll expand persona count only if time allows in later sprints

3-Sprint Roadmap

Sprint 1 — Design & Bootstrap: Infrastructure is live (Next.js frontend on Vercel, Firebase Auth/Firestore connected, Python FastAPI persona service deployed and reachable), and the office-floor game world plus one persona's dialogue design are complete as a prototype, no working features yet.

Sprint 2 — Raw MVP: The core loop's main feature; a player can walk up to one LLM-powered client persona and run a cold-outreach and live-meeting exchange that responds dynamically; is built end-to-end, even if rough.

Sprint 3 — Refine MVP: Additional personas, portfolio/progress tracking, and UI polish are added; the build is playtested with IBM consultants for authenticity and hardened toward presentable, closer-to-production quality.

Success Criteria

By the end of the term, a player with no consulting background can complete a full engagement loop unaided; finding a lead, running cold outreach, prepping for and holding a client meeting, and closing a contract; entirely within the game, with at least one persona responding dynamically rather than through fixed dialogue trees, and with mechanics that trace back to real practice shared by IBM consultants.

Assumptions & Constraints

- No IBM or RMIT-provided API/LLM credits; the team is funding its own Anthropic API usage for persona dialogue (estimated well under \$50 for the term at expected usage volumes)
- IBM's own stack (watsonx, Orchestrate, governance) is confirmed optional; not required for this build
- Access to practising IBM consultants for interviews/shadowing and later authenticity playtesting is assumed but not yet scheduled
- This proposal is pending formal client sign-off at our next meeting with Emily

Client Agreement

Both the client and a team representative should confirm this proposal reflects what was discussed before Part B is built out. To be completed at the follow-up client meeting.

Client name:

Client signature / date:

Team representative: Gayath

Team signature / date: 13/08/2026

Part B — Sprint 1 Plan (Weeks 1–3)

Every team member contributes 4 hours/week. Tasks sit on the Infrastructure track or the Design/Product track no working features yet on either track this sprint.

Week 5

Stand up Next.js + Firebase (Auth/Firestore) and deploy to Vercel

Owner	Kashaf Fatima	Track	Infrastructure
Week	5	Hours	4

Description: Provision the Next.js app on the Firebase project, connect Firestore and Auth, and deploy to Vercel so a placeholder page is live at a real URL.

Deliverable: Live Vercel URL with Firestore/Auth connected.

Acceptance criteria:

- ✓ App deployed and reachable at a real URL
- ✓ Firestore and Auth confirmed connected
- ✓ Placeholder route returns a successful response

Scaffold the FastAPI persona service and deploy it

Owner	Ibrahim Allouche	Track	Infrastructure
Week	5	Hours	4

Description: Bootstrap a Python FastAPI service, add a placeholder /health endpoint, and deploy it to Render/Railway so it's reachable from the internet.

Deliverable: Live FastAPI URL responding to /health.

Acceptance criteria:

- ✓ Service provisioned and running
- ✓ /health endpoint returns a successful response
- ✓ Deployed URL shared with the team

Wire Next.js → FastAPI communication and shared env config

Owner	Ibrahim Allouche	Track	Infrastructure
Week	5	Hours	4

Description: Confirm the Next.js frontend can call the FastAPI /health endpoint end-to-end, and document the environment variables both services need.

Deliverable: Working request/response between frontend and backend; env var reference doc.

Acceptance criteria:

- ✓ Frontend successfully calls backend and displays the response
- ✓ Env vars documented for both services
- ✓ No secrets committed to Git

Draft initial requirements from client discussion

Owner	Fatima Hubali	Track	Design / Product
Week	5	Hours	4

Description: Turn the 7 Aug client discussion and project brief into a short requirements document covering the problem, target users, and the main feature Sprint 2 will need to prove.

Deliverable: Requirements document shared with the team.

Acceptance criteria:

- ✓ Problem and target users documented
- ✓ Candidate main feature for Sprint 2 identified
- ✓ Shared with client for confirmation at the next meeting

Prepare consultant interview/shadowing plan

Owner	Gayath Wethmin	Track	Design / Product
Week	5	Hours	4

Description: Draft an interview guide for shadowing a practising IBM consultant, covering lead-finding, cold outreach, meeting prep, and live client engagement, and coordinate scheduling with the client contact.

Deliverable: Interview question guide + scheduled (or requested) session.

Acceptance criteria:

- ✓ Interview guide covers all four lifecycle stages
- ✓ Session requested or scheduled with client
- ✓ Guide shared with BA/UX for input

Mock up the office-floor game world and first dialogue UI

Owner	Amritha Selvaganapathi	Track	Design / Product
Week	5	Hours	4

Description: Produce a Figma mockup of the office-floor setting, the player/NPC sprite style direction, and the dialogue box UI triggered when the player walks up to an NPC.

Deliverable: Figma file with office scene + dialogue UI mockup.

Acceptance criteria:

- ✓ Office scene layout mocked up
- ✓ Dialogue box UI mocked up
- ✓ Style direction agreed with the team

Week 6

Design Firestore schema for session/persona state

Owner	Kashaf Fatima	Track	Infrastructure
Week	6	Hours	4

Description: Define Firestore collections for sessions, personas, and portfolio progress, and implement default-deny security rules.

Deliverable: Schema doc + deployed Firestore rules.

Acceptance criteria:

- ✓ Collections defined for sessions, personas, progress
- ✓ Security rules deployed
- ✓ Schema documented for the team

Build first Phaser scene: office tilemap + player movement

Owner	Ibrahim Allouche	Track	Infrastructure
Week	6	Hours	4

Description: Embed a Phaser 3 canvas in a Next.js page, render the office tilemap from the UX mockup, and implement player movement with collision.

Deliverable: Playable local build with a walkable office scene.

Acceptance criteria:

- ✓ Tilemap renders correctly
- ✓ Player character moves with collision
- ✓ Scene runs inside the Next.js page

Scaffold persona prompt handling in FastAPI

Owner	Kashaf Fatima	Track	Infrastructure
Week	6	Hours	4

Description: Implement a first system-prompt template and an Anthropic API call in FastAPI that returns a persona-style response to a test message.

Deliverable: Working /persona/respond endpoint calling the Anthropic API.

Acceptance criteria:

- ✓ Endpoint accepts a message and returns a persona-style reply
- ✓ API key handled via env var, not committed
- ✓ Response verified against a manual test case

Draft first client persona brief and dialogue objections

Owner	Fatima Hubali	Track	Design / Product
Week	6	Hours	4

Description: Turn consultant interview/shadowing notes into a persona brief — personality, goals, and realistic objections — for the first LLM-powered client.

Deliverable: Persona brief document.

Acceptance criteria:

- ✓ Personality and goals defined
- ✓ At least 3 realistic objections captured
- ✓ Grounded in interview/shadowing notes, not invented

Paper prototype the full engagement loop flow

Owner	Amritha Selvaganapathi	Track	Design / Product
Week	6	Hours	4

Description: Map the end-to-end flow — find a lead, cold outreach, meeting prep, live meeting, close — as a Figma flow/paper prototype.

Deliverable: Figma flow covering the full loop.

Acceptance criteria:

- ✓ All four lifecycle stages represented
- ✓ Screen-to-screen flow is walkable in Figma
- ✓ Reviewed with BA for accuracy against real practice

Confirm Sprint 2 scope with client and supervisor

Owner	Gayath Wethmin	Track	Design / Product
Week	6	Hours	4

Description: Hold the client follow-up meeting to formally agree Part A, confirm the Sprint 2 main feature, and check in with the supervisor.

Deliverable: Signed-off Part A; meeting notes.

Acceptance criteria:

- ✓ Client has confirmed the proposal reflects the discussion

- ✓ Sprint 2 main feature explicitly agreed
- ✓ Supervisor meeting held before end of Week 4

Week 7

Connect NPC walk-up trigger to the persona endpoint end-to-end

Owner	Ibrahim Allouche	Track	Infrastructure
Week	7	Hours	4

Description: Wire the Phaser walk-up-to-NPC trigger to call the FastAPI /persona/respond endpoint and display the reply in the dialogue UI.

Deliverable: End-to-end round trip: walk up to NPC → live LLM reply shown in dialogue box.

Acceptance criteria:

- ✓ Walking into an NPC opens the dialogue UI
- ✓ A live Anthropic API response is displayed
- ✓ Round trip verified with a manual playtest

Firebase Auth flow and session persistence

Owner	Kashaf Fatima	Track	Infrastructure
Week	7	Hours	4

Description: Gate the game world behind login and persist session state (player progress) to the Firestore session document defined in Week 2.

Deliverable: Working login → game world flow with persisted session doc.

Acceptance criteria:

- ✓ Login required before entering the game world
- ✓ Session document created/updated in Firestore
- ✓ Reload preserves session state

Finalize Sprint 1 requirements doc and confirm with client

Owner	Fatima Hubali	Track	Design / Product
Week	7	Hours	4

Description: Incorporate feedback from the Week 2 client meeting into the requirements document and circulate the final version.

Deliverable: Finalized requirements document.

Acceptance criteria:

- ✓ Feedback from client meeting incorporated

- ✓ Document finalized and shared with the team
- ✓ Confirmed with client

UI polish pass on dialogue box and office scene assets

Owner	Amritha Selvaganapathi	Track	Design / Product
Week	7	Hours	4

Description: Apply the agreed style direction to the dialogue box and office scene assets so Sprint 1's prototype looks presentable for the demo.

Deliverable: Updated Figma assets applied to the build.

Acceptance criteria:

- ✓ Dialogue box matches agreed style direction
- ✓ Office scene assets updated
- ✓ Assets handed off to Dev for integration

Prepare and run Sprint 1 review with client and supervisor

Owner	Gayath Wethmin	Track	Design / Product
Week	7	Hours	4

Description: Prepare a short walkthrough of the Sprint 1 deliverables (live infra + design prototype) and present it to the client and supervisor.

Deliverable: Sprint 1 review meeting held; notes and any follow-up actions recorded.

Acceptance criteria:

- ✓ Review meeting held before end of Sprint 1
- ✓ Live infra and prototype demonstrated
- ✓ Feedback captured for Sprint 2 planning